Randomised Complete Block Design

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Introduction

A randomised complete block design (RCBD) is one of the most useful experimental designs in applied research. When a known source of variation – day of measurement, animal cage, plate, subject – is likely to influence the outcome, RCBD groups experimental units into homogeneous blocks and randomises treatments *within* each block. The block effect is thereby separated from the treatment effect, and the residual error is the within-block variability, which is typically much smaller than the pooled unblocked error.

Prerequisites

The reader should have met the one-way ANOVA and the principle of randomisation. Familiarity with R's formula syntax is assumed.

Theory

Let y_{ij} be the response of the experimental unit receiving treatment i in block j, for $i = 1, \ldots, t$ treatments and $j = 1, \ldots, b$ blocks. The additive RCBD model is

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2),$$

where τ_i is the treatment effect (summing to zero) and β_j is the block effect. Each treatment appears exactly once in each block – that is the "complete" in RCBD.

The ANOVA decomposes the total sum of squares into treatment, block, and residual components. The F-test for treatments has $t - 1$ numerator and $(t - 1)(b - 1)$ denominator degrees of freedom. If blocking has captured real variation, the residual mean square is smaller than it would have been in a completely randomised design, and the test has higher power.

The design assumes **no treatment-by-block interaction**. If an interaction exists (a treatment works better in some blocks than others), the additive model is misspecified. The Tukey one-degree-of-freedom test for non-additivity provides a partial check.

Assumptions

1. **Additivity**: no treatment-by-block interaction.
2. **Independence**: residuals are mutually independent.
3. **Homoscedasticity**: residuals have constant variance across treatments and blocks.
4. **Normality**: residuals are approximately normal.

R Implementation

```r
library(agricolae)
library(emmeans)

treatments <- LETTERS[1:4]
blocks <- paste0("B", 1:5)
```

```r
set.seed(2026)
design <- design.rcbd(trt = treatments, r = length(blocks),
                      seed = 2026)
print(design$book)

design$book$y <- with(design$book,
                      5 +
                        as.numeric(factor(block)) * 0.8 +
                        c(0, 1.2, 1.8, 0.4)[as.numeric(factor(treatments))] +
                        rnorm(nrow(design$book), 0, 0.6))

fit <- aov(y ~ block + treatments, data = design$book)
summary(fit)

emm <- emmeans(fit, ~ treatments)
pairs(emm, adjust = "tukey")
```

`design.rcbd()` produces a randomised plan with each treatment appearing once per block. The simulated response includes a block effect and a treatment effect with the lowest for A, highest for C. `aov()` fits the additive RCBD model; `emmeans::pairs()` performs pairwise comparisons with Tukey adjustment.

Output & Results

The ANOVA table:

```
            Df Sum Sq Mean Sq F value  Pr(>F)
block        4 18.120   4.530  12.815 0.00015 ***
treatments   3  9.452   3.151   8.916 0.00142 **
Residuals   12  4.242   0.353
```

The block effect accounts for a substantial fraction of the variability – confirming that blocking was worthwhile – and the treatment effect is significant at $p = 0.0014$. Pairwise comparisons show C significantly higher than A ($p < 0.001$) and B ($p = 0.03$), and D not significantly different from A.

Interpretation

For a manuscript: "Treatments were compared in a randomised complete block design with four treatments and five blocks. An additive ANOVA showed a significant treatment effect, F(3, 12) = 8.92, $p = 0.001$, after accounting for block variation (which itself was substantial, F(4, 12) = 12.82, $p < 0.001$). Tukey-adjusted pairwise comparisons indicated that treatment C produced significantly higher responses than treatments A and B (see Table X)."

Practical Tips

- Block on variables that you expect to influence the outcome and that you cannot randomise out. Common choices: day of measurement, microtitre plate, cage, field plot, subject.
- Do not include the block as a "predictor of interest" – its role is to remove nuisance variation. Reporting its effect is informative but rarely the headline.
- If you suspect treatment-by-block interaction, switch to a design that models it: a randomised complete block with replicates within blocks, or a factorial design with block as a factor.
- When the number of treatments exceeds what fits in a block (e.g., 8 treatments but only 4 samples per plate), use an incomplete block design (BIBD).
- Always include randomisation within each block – the "R" in RCBD is essential for validity, even though the formula structure looks the same.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans